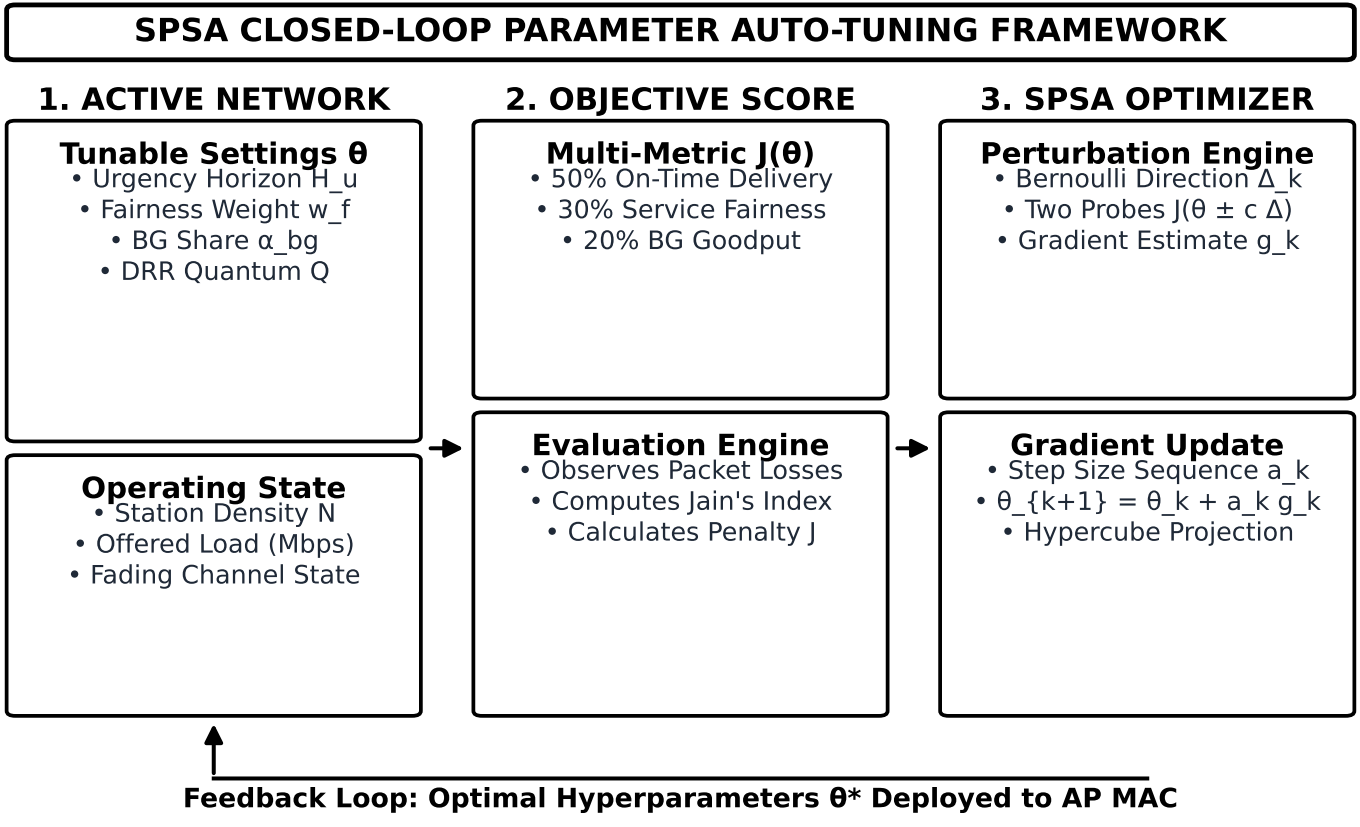
SPSA CLOSED-LOOP PARAMETER AUTO-TUNING FRAMEWORK
1. ACTIVE NETWORK
Tunable Settings θ
• Urgency Horizon H_u
• Fairness Weight w_f
• BG Share α_bg
• DRR Quantum Q
Operating State
• Station Density N
• Offered Load (Mbps)
• Fading Channel State
2. OBJECTIVE SCORE
Multi-Metric J(θ)
• 50% On-Time Delivery
• 30% Service Fairness
• 20% BG Goodput
Evaluation Engine
• Observes Packet Losses
• Computes Jain's Index
• Calculates Penalty J
3. SPSA OPTIMIZER
Perturbation Engine
• Bernoulli Direction Δ_k
• Two Probes J(θ ± c Δ)
• Gradient Estimate g_k
Gradient Update
• Step Size Sequence a_k
• θ_{k+1} = θ_k + a_k g_k
• Hypercube Projection
Feedback Loop: Optimal Hyperparameters θ* Deployed to AP MAC